BEU CONNECT

Bihar Engineering University, Patna

B.Tech 2nd Semester Examination - 2026

Time: 03 Hours

Code: 100215 ***Subject: Engineering Chemistry*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2×7=14]**

(a) The shape of the ammonia (NH_3) molecule according to VSEPR theory is?
- (i) Trigonal planar
- (ii) Linear
- (iii) Trigonal pyramidal
- (iv) Tetrahedral

(b) Which of the following metal carbonyls obeys the 18-electron rule?
- (i) $Fe(CO)_5$
- (ii) $Ni(CO)_4$
- (iii) $Cr(CO)_4$
- (iv) $Mn(CO)_2$

(c) UV-Visible spectroscopy mainly involves transitions of
- (i) Nuclear spins
- (ii) Core electrons only
- (iii) Protons only
- (iv) Valence electrons

(d) The resonance frequency, in NMR spectroscopy depends mainly on
- (i) Molecular weight
- (ii) External magnetic field
- (iii) Density
- (iv) Melting point

(e) The Nernst equation is used to determine
- (i) Rate of reaction
- (ii) Heat of reaction
- (iii) Molecular weight
- (iv) Cell potential under non-standard conditions

(f) Which electrolyte is used in a lead-acid battery?
- (i) NaOH
- (ii) Dilute H_2SO_4
- (iii) HCl
- (iv) KOH

(g) Temporary hardness of water is mainly due to
- (i) Calcium chloride
- (ii) Magnesium sulfate
- (iii) Sodium chloride
- (iv) Calcium bicarbonate

(h) Which process removes hardness by exchanging Ca^{2+} and Mg^{2+} ions with Na^{+} ions?
- (i) Lime-soda process
- (ii) Zeolite process
- (iii) Distillation
- (iv) Reverse osmosis

(i) Which polymer is an example of a thermoplastic?
- (i) Bakelite
- (ii) Melamine
- (iii) Polyethylene
- (iv) Epoxy resin

(j) Silicones are classified as
- (i) Natural polymers
- (ii) Condensation monomers
- (iii) Biopolymers
- (iv) Inorganic polymers

Q.2

(a) Discuss the Photoelectric Effect and Heisenberg's Uncertainty Principle with suitable examples. [7]

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store

(b) Describe the formation of Molecular Orbitals using the Linear Combination of Atomic Orbitals. Draw and explain the molecular orbital diagram of O_2. [7]

Q.3

(a) Explain the principles of rotational and vibrational spectroscopy and applications of these techniques in structural elucidation of organic molecule. [7]

(b) Describe the principle, instrumentation, and applications of UV-Visible spectroscopy in chemistry. [7]

Q.4

(a) Derive the Nernst equation and explain its significance in determining the EMF of electrochemical cells under non-standard conditions. [7]

(b) Explain the mechanism of corrosion. Discuss different types of corrosion with suitable examples. [7]

Q.5

(a) What is hardness of water? Explain the types of hardness and describe the estimation of hardness by the EDTA method. [7]

(b) Discuss in details the Lime-Soda process for the removal of hardness of water. [7]

Q.6

(a) Differentiate between thermoplastics and thermosetting plastics. Discuss the preparation, properties and applications of commercially important thermoplastics such as Polyethylene, Polyvinyl Chloride and Polystyrene. [7]

(b) Describe the preparation, properties and applications of Phenol-Formaldehyde resin (Bakelite). [7]

Q.7

(a) Explain the Diels–Alder cyclization reaction and epoxide ring-opening reaction and their applications in organic synthesis. [7]

(b) Discuss the synthesis of aspirin from salicylic acid. Explain the reaction mechanism and significance of aspirin as an important pharmaceutical drug. [7]

Q.8 Write Short note on:

(a) Chemical Shift in NMR spectroscopy. [7]

(b) Working and advantages of fuel cells. [7]

Q.9 Write short note on any two of the following: **[7×2=14]**

(a) Bonding metal carbonyls

(b) Boiler problems such as sludge formation, scale formation, priming and foaming

(c) Oxidation-reduction reactions in organic chemistry

(d) Mechanism of free radical polymerization reaction

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store